

GPU based Face Verification under Surveillance Scenario

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Abstract

Face Verification (FV) is the most accepted method of biometric authentication due to its inherent passive nature. This has attracted a lot of researchers over past few decades to achieve an moderately high accuracy under controlled environments. In order to achieve such an accuracy for FV under surveillance scenario has been proved to be a major hurdle in this area of research, mainly due to the difference in resolution, contrast, illumination and camera parameters of the training and the testing samples. Deep Learning techniques based on Convolutional Neural Network has been used in the recent past to cope with the problem of FR, with much success in the controlled environments. We aim to bridge the gap in the distributions of the gallery and probe samples, based on their convolutional feature, using Domain Adaptation techniques. The convolutional feature extraction are very expensive in computationally, hence they need to be parallelised on GPU for efficient training.

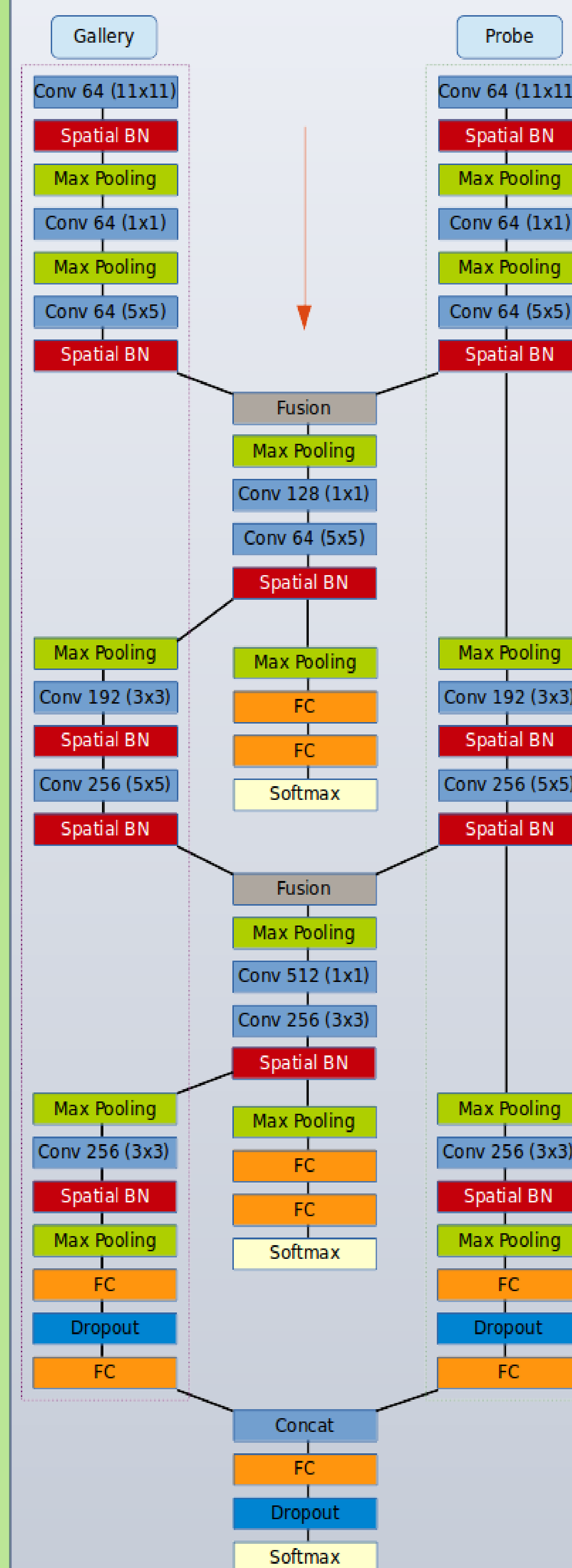
Introduction

Face Verification (FV) is a classical problem which is far from being solved. Face Verification has a clear advantage of being natural and passive over other biometric techniques requiring co-operative subjects. Most face recognition algorithms perform well under a controlled environment. Face images obtained by an outdoor panoramic surveillance camera, are often confronted with severe degradations (e.g., low-resolution, blur, low-contrast, interlacing and noise). This significantly limits the performance of face recognition systems used for binding “security with surveillance” applications. Here, images used for training are usually available beforehand which are taken under a well controlled environment in an indoor setup (laboratory, control room), whereas the images used for testing are captured when a subject comes under a surveillance scene. Performance of most classifiers degrade when both the resolution and contrast of face templates used for recognition are low. For surveillance applications, a face recognition system must recognize a face in an unconstrained environment without the notice of the subject. Degradation of faces is quite evident in the surveillance scenario due to low-resolution and camera-blur. Variations in the illuminating conditions of the faces not only reduces the recognition accuracy but occasionally degrades the performance of face detection which is the first step of face recognition.

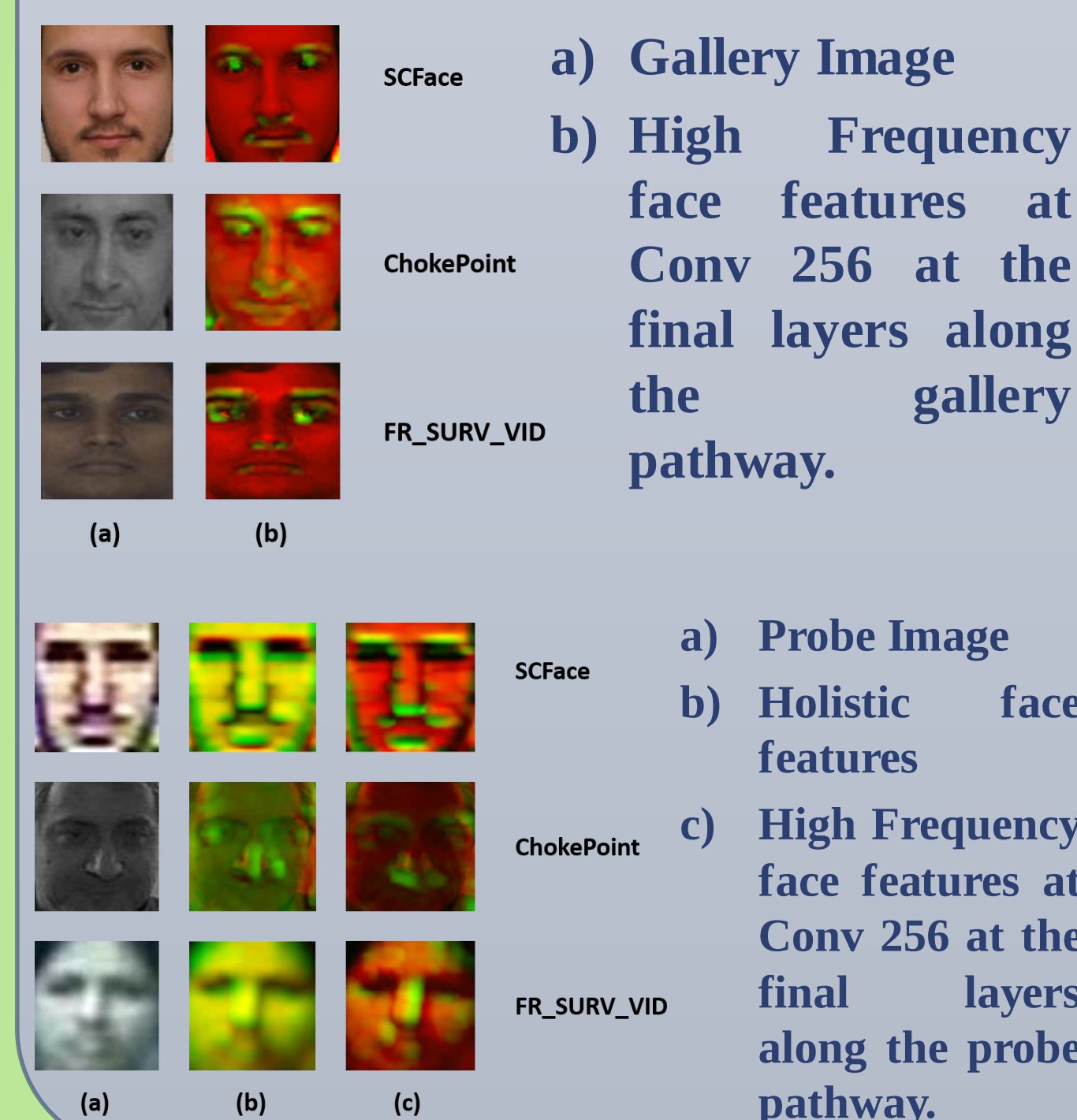
Objectives

- ❖ Face Detection using *Chehra* face detector.
- ❖ Data Augmentation.
- ❖ Pre-processing of the tightly cropped image:
 - ❖ Downsampling of the gallery face images.
 - ❖ Face Hallucination (upsampling) of the probe face images.
 - ❖ Degradation (blur) of the gallery samples using Gaussian blur.
 - ❖ Contrast enhancement of the probe samples.
- ❖ Feature extraction using a Deep Convolutional network.
- ❖ Domain Adaptation fusion based CNN.
- ❖ Multiple interactive-fusion to reduce the disparity between the face images from gallery and probe.

FACE FUSION NETWORK



Results and Discussion



Dataset

Dataset	Training	Testing
SCFace	2000	3000
ChokePoint	5000	10000
FR_SURV_VID	4000	10000



- Dual Pathway based CNN.
- FV based on multiple interaction.
- At each interaction the disparity between the gallery and probe is evaluated as SOFTMax loss.
- Fully-connected or the FC layers are the specialized layers, which focusses on the high-frequency elements of the face.
- The convolutional layers are stacked over *Rectified Linear Unit (ReLU)* and a *MAXPOOLING* layer.
- ReLU induces non-linearity and sparsity in the network, due to its activation function $h = \max(0, a)$ where $a = Wx + b$, W and b are the weights and bias of the layer.
- Max-pooling is a form of non-linear downsampling, which helps to eliminate the non-maximal values to reduce the computation of the higher layers and provides a form of translational invariance.
- As the data flows through a deep network, the weights and parameters adjust those values, sometimes making the data too big or too small, referred to as “internal covariate shift”. By normalizing the data in each mini-batch, this problem is largely avoided. The batch-normalization layer is adapted from the method proposed by Ioffe and Szegedy.

Accuracy

Algorithm	SCface	CP	FSV
EDA	46.35	75.21	32.52
DeepFace	52.67	76.26	40.32
SML_MFKC	57.51	76.23	54.62
FaceNET	69.08	83.72	54.68
Face Fusion	72.26	84.12	56.45

Conclusion

- ❖ The Face fusion technique performs better than all other faces under comparison.
- ❖ The number of parameters of the Face fusion network ~210M.
- ❖ CPU running time per epoch (Intel core i7-4720K) ~ 200 hrs.
- ❖ GPU running time per epoch (dual GTX Titan X) ~2 hrs.
- ❖ Deep learning techniques not only outperforms shallow techniques, but our method of fusion outperforms all other methods by a considerable margin.

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